

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Researching the security of an LLM-controlled robot, where the cheap per-step gate screening its motion plans is the exact seam an attacker slips a harmful plan through.
- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, the gate the decomposition attack is built to defeat.
- Architected the node graph to run byte-identical in Gazebo and on a physical RoboMaster EP Core, isolating the sim-to-hardware swap to the URDF and controller layer.

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Spring Boot, Neo4j, and PostgreSQL stack.
- Authored an AWS deployment script and managed the environment as code with Terraform, replacing a manual multi-service bring-up that took 30 minutes with a single-command, version-controlled provision so the polyglot stack deploys identically every time.

Robostangs (FRC Team 548)

Software Captain

Jun 2023 – May 2024

Northville, MI

- Led the software and electrical teams (5 and 6 engineers) for a competition robot that reached the FIRST World Championship and placed top 16 in its division in Houston, the program's strongest world-stage finish since 2014.
- Engineered graceful sensor-failure handling so the most match-critical mechanism falls back to an internal encoder and stays operable through a sensor loss instead of bricking mid-match, surfacing the degradation as an alert rather than a silent failure.

PROJECTS

Dadei | *Python, FastAPI, React, Redis, PostgreSQL, Docker*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Engineered a no-merge-first speaker-identification model using an EMA voice centroid plus a bounded prototype bank with floor-and-margin gating, abstaining into a new identity on ambiguity rather than collapsing two people into one, attributing 96% of utterances to the correct speaker.
- Built a React client that streams microphone audio over a WebSocket with client-side Web Audio processing, rendering live captions in under 300ms as the backend transcribes.

fliks | *Go, Docker, PostgreSQL*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Chose a single Go monolith over a microservice split, keeping auth, rate-limiting, and rating aggregation in-process because premature decomposition is the harder call to defend at pre-scale, one deploy unit against one datastore.
- Built a self-hosted video transcoding pipeline with a concurrent Go worker that pulls upload jobs, transcodes, and writes processed video back to object storage, owning the media path end to end instead of renting a managed service.

TECHNICAL SKILLS

Languages Python, Java, Go, C++, JavaScript/TypeScript, SQL, Bash
Frameworks FastAPI, Spring Boot, React
Robotics ROS2, Gazebo
Cloud & Infrastructure AWS (EC2, S3, RDS), Docker, Terraform, Redis